A FINITE FLAT MODULE THAT IS NOT PROJECTIVE, FROM A RING OF SMOOTH FUNCTIONS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file examples.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: Both counted items are witnessed by the single displayed example; the delegation target is Algebra, Remark warning, which resolves inside the corpus pool as with the other Algebra references in this chapter.

Editorial note. The label lemma-finite-flat-non-projective names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Context span 2622-2632: the section declares itself a copy of Algebra, Remark warning (intra-corpus reference); the counterexample data: R = C^infinity(R) the ring of infinitely differentiable functions on R, m = {f in R : f(0) = 0}, I = {f in R : there exists epsilon > 0 with f(x) = 0 for all |x| < epsilon}, and M = R_m = R/I
- Context span 2634-2636: the morphism Spec(R/I) -> Spec(R) is also an example of a flat closed immersion which is not open

Source reference. Original remark 00NY: sources/stacks-chow/tags.txt, source lines 876–876. Prerequisite pointer invoked at 2653; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

This is a copy of Algebra, Remark 00NY. It is not true that a finite R-module which is R-flat is automatically projective. A counter example is where $R = \mathcal{C}^\infty(\mathbf{R})$ is the ring of infinitely differentiable functions on $\mathbf{R}$, and $M = R_{\mathfrak{m}} = R/I$ where $\mathfrak{m} = \{f \in R \mid f(0) = 0\}$ and $I = \{f \in R \mid \exists \epsilon, \epsilon > 0 : f(x) = 0 \ \forall x, |x| < \epsilon\}$.

The morphism $\mathrm{Spec}(R/I) \to \mathrm{Spec}(R)$ is also an example of a flat closed immersion which is not open.

Lemma 0.1. *Strange flat modules.*

(1) *There exists a ring R and a finite flat R-module M which is not projective.*
(2) *There exists a closed immersion which is flat but not open.*

Proof. See discussion above. □

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